FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Advanced)-2020

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B.**

1. ***Section-A (01– 7, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section-A (08 – 13, 26 – 31, 44 – 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S). Each multiple choice question with 4 options has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section–B (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answers as numerical value from 0 to 9 and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. An L-C-R series circuit with 100 Ω resistance is connected to an *ac* source. When only the capacitance is removed, the current lags behind voltage by $\frac{\pi}{4}$. When only the inductance is removed, the current leads voltage by $\frac{\pi}{4}$. Then
   (A) Inductive reactance is 100 Ω
   (B) Capacitive reactance is 100 Ω
   (C) Impedance of L-C-R circuit is 100 Ω
   (D) Maximum potential difference across inductor is equal to maximum potential difference across capacitor when all are connected in circuit.

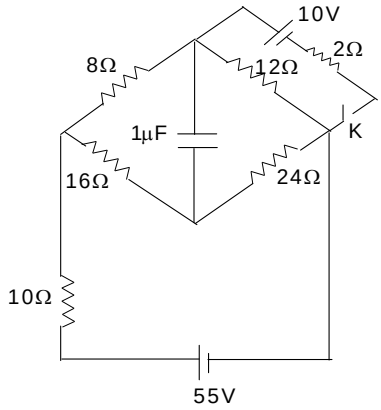
2. When a monochromatic point source of light is at a distance of 0.2 m from a photoelectric cell, the cut-off voltage and the saturation current are respectively 0.6 V and 18.0 mA. If the same source is placed 0.6 m away from the photoelectric cell, then
   (A) the stopping potential will be 0.2 V
   (B) the stopping potential will be 0.6 V
   (C) the saturation current will be 6.0 mA
   (D) the saturation current will be 2.0 mA

3. A wave disturbance in a medium is described by $y(x,t) = 0.02\cos\left(50\pi t + \frac{\pi}{2}\right)\cos(10\pi x)$
   where *x* and *y* are in meter and *t* is in second. Then
   (A) First node occurs at *x* = 0.15m
   (B) First antinode occurs at *x* = 0.3 m
   (C) The speed of interfering waves is 5.0 m/s
   (D) The wavelength is 0.2 m

4. A planet of mass *m* moves along an ellipse around the sun of mass *M* (the sun is at one focus) so that its maximum and minimum distances from the sun are r_1 and r_2 and velocities at these points are v_1 and v_2 respectively. Then
   (A) $v_1 r_1 = v_2 r_2$
   (B) total energy of planet is always constant
   (C) angular momentum of planet is $m\sqrt{\frac{2GM(r_1 r_2)}{(r_1 + r_2)}}$
   (D) $v_2 r_1 = v_1 r_2$

Space for Rough work

5. In the given circuit,



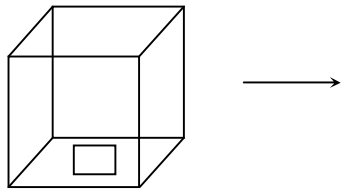
(A) current flowing through the 10Ω resistor is 2.36 A when key is open

(B) Net charge on capacitor is 4.8 μc when key is open

(C) Net charge on capacitor is 4.8 μc when key is closed

(D) current flowing through 10Ω resistor is 2.7 A, when key is closed

6. A solid block of mass 2 kg is resting inside a cube as shown in the figure. The cube is moving with the velocity $\vec{v} = 5t\hat{i} + 2\hat{j}$ m/s (*t* is time in second). The block is at rest with respect to the cube and coefficient of friction between the surfaces of cube and the block is 0.6. Then (g = 10 m/s^2)



(A) force of friction acting on the block is 10 N.

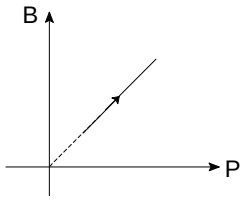
(B) force of friction acting on the block is 4 N.

(C) the total force exerted by the block on the cube is 14 N.

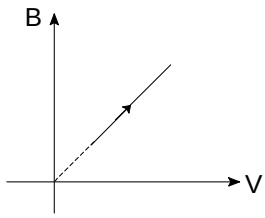
(D) the total force exerted by the block on the cube is $10\sqrt{5}$ N.

7. A sample of gas follow process represented by PV^2 = constant. Bulk modulus for this process is B. Then which of the following graph is correct?

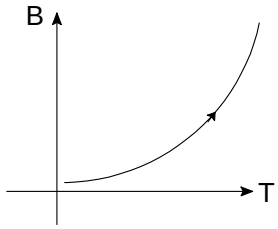
(A)



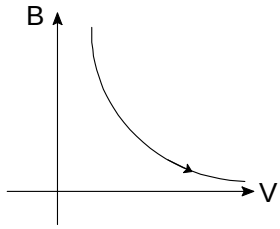
(B)



(C)



(D)



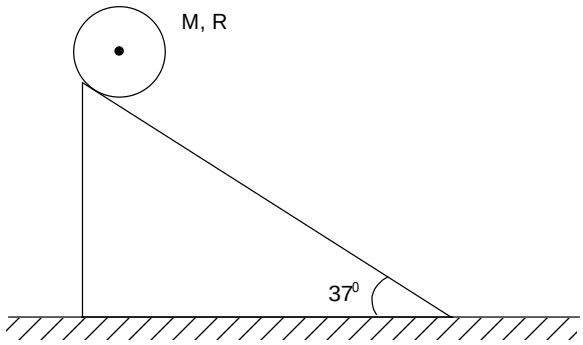
Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.

A body of mass M and radius R released from the top of inclined plane of length L. If friction is sufficient to prevent slipping. Then column – 1 represent shape of rigid body, column – 2 represent friction acting on rigid body and column – 3 time to reach of bottom of Incline plane. Answer the following:-



Column – 1		Column – 2		Column – 3	
(I)	Solid sphere	(i)	$\frac{mg}{5}$	(P)	$\sqrt{\frac{20L}{3g}}$
(II)	Disc	(ii)	$\frac{6mg}{35}$	(Q)	$\sqrt{\frac{50L}{9g}}$
(III)	Ring	(iii)	$\frac{6mg}{25}$	(R)	$\sqrt{\frac{5L}{g}}$
(IV)	Hollow sphere	(iv)	$\frac{3mg}{10}$	(S)	$\sqrt{\frac{14L}{3g}}$

8. Choose the correct alternative which will take maximum time to reach at bottom of incline

(A) $(I)(ii)(P)$ (B) $(II)(iv)(P)$

(C) $(IV)(i)(P)$ (D) $(III)(iv)(P)$

Space for Rough work

9. Combination which takes minimum time to reach at the bottom of incline
(A) $(II)(i)(S)$ (B) $(IV)(iii)(S)$
(C) $(I)(ii)(S)$ (D) $(III)(iv)(S)$

10. Pick the correct combination
(A) $(II)(i)(Q)$ (B) $(IV)(iii)(R)$
(C) $(IV)(iii)(Q)$ (D) $(II)(iii)(R)$

Answer questions 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.

A charged particle (elelctron or proton) is introduced at the origin (0, 0, 0) with a given initial velocity $\vec{v}$, electric field $\vec{E}$ and magnetic field $\vec{B}$ are given in column I, 2, 3 respectively. The quantities E_0, B_0 are +ve in nature.

Column – 1		Column – 2		Column – 3	
(I)	Electron with $\vec{V} = 2\frac{E_0}{B_0}\hat{i}$	(i)	$\vec{E} = E_0\hat{K}$	(P)	$\vec{B} = -B_0\hat{i}$
(II)	Electron with $\vec{V} = \frac{E_0}{B_0}\hat{j}$	(ii)	$\vec{E} = -E_0\hat{j}$	(Q)	$\vec{B} = B_0\hat{i}$
(III)	Proton with $\vec{V} = 0$	(iii)	$\vec{E} = -E_0\hat{i}$	(R)	$\vec{B} = B_0\hat{j}$
(IV)	Proton with $\vec{V} = \frac{2E_0}{B_0}\hat{i}$	(iv)	$\vec{E} = E_0\hat{i}$	(S)	$\vec{B} = B_0\hat{k}$

11. In which case particle will move in a straight line with constant velocity.
(A) $(II)(iii)(S)$ (B) $(IV)(i)(S)$
(C) $(III)(ii)(R)$ (D) $(III)(iii)(P)$

Space for Rough work

12. In which case particle describe a helical path along +Z direction.
(A) (II)(ii)(R) (B) (IV)(ii)(R)
(C) (IV)(i)(S) (D) (III)(iii)(P)

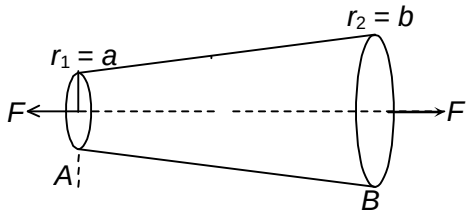
13. In which case, would the particle move in a straight line along –ve direction of y – axis.
(A) (IV)(ii)(S) (B) (III)(ii)(P)
(C) (II)(iii)(Q) (D) (III)(ii)(R)

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

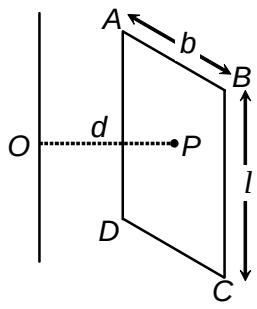
14. An infinitesimally small bar magnet of dipole moment M =1 A-m^2 is pointing and moving with the speed v = 1m/s in the x-direction. A small closed circular conducting loop of radius a = 1cm and of negligible self-inductance lies in the y-z plane with its centre at x = 0, and its axis coinciding with the x-axis. The force opposing the motion of the magnet when x = 1m is found to be $4y \times 10^{-22}$ N. The resistance of the loop $R = 10\Omega$. Find the value of y.

15. A slightly tapering wire of length L = 8 m and end radii a = 1 mm and b = 4 mm is subjected to stretching forces $F = 10^3$N as shown in Figure. If $Y = \pi \times 10^{11}$ N/m^2 is Young's modulus, calculate the extension produced in the wire. ($\pi^2 = 10$)

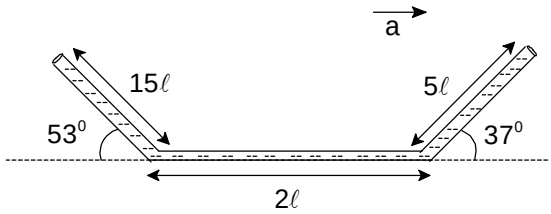


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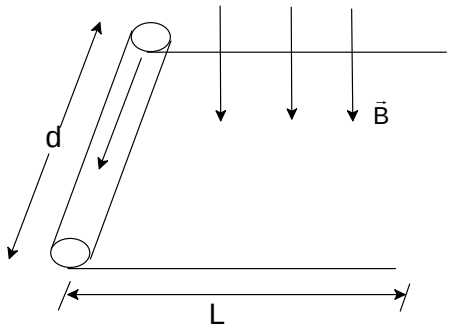
16. Find the electric flux (in SI unit) crossing the wire frame *ABCD* of length *l* = 1m width *b* and whose center is at a distance *OP* = *d* (=*b*/2) from an infinite line of charge with linear charge density $\lambda = \pi/18 \times 10^{-9}$ c/m. Consider that the plane of frame is perpendicular to line *OP*.



17. In the given tube a liquid of density ρ is filled. If the acceleration of the tube towards right direction such that no liquid will falls out of the tube is $\frac{\eta g}{5}$. Find the value of η



18. A rod of mass M = 3 kg and radius 5 cm rests on two parallel rails that are d (10 cm) apart and L (2 m) long. The rod carries a current I = 50A (in the direction shown) and rolls along the rails without slipping. If it starts from rest at one end. And a uniform magnetic field of magnitude 0.4T is directed perpendicular to the rods and the rails. It leaves the rail at a speed of $\frac{K}{3}$ m / s. Find K.



Space for Rough work

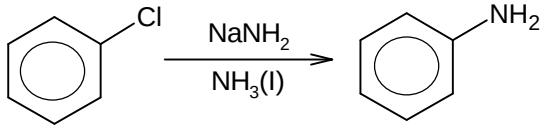
Chemistry

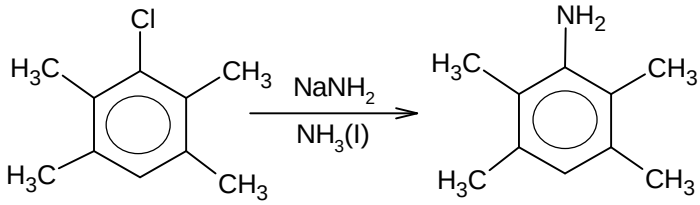
PART – II

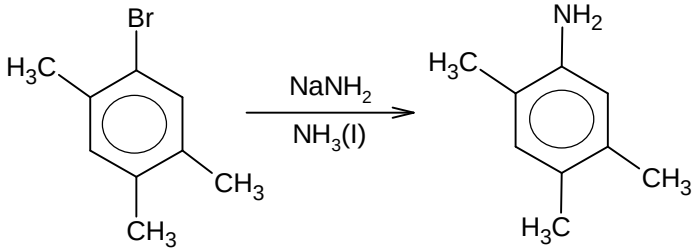
SECTION – A
(One or More than one correct type)

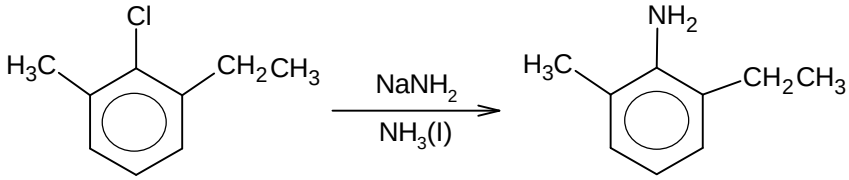
This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Which of the following reaction(s) is/are not feasible?

(A)


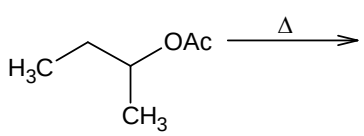
(B)


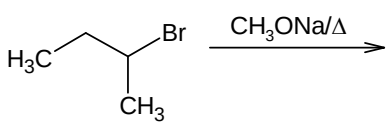
(C)


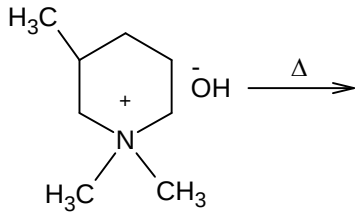
(D)


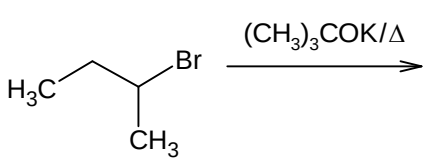
Space for Rough work

20. In which of the following reaction(s), product formation takes place by Hofmann rule?

(A) 

(B) 

(C) 

(D) 

21. An electrochemical cell is made by combining two half cells
(X) $AgNO_3$(0.05) M) | Ag(s) and
(Y) $AgNO_3$ (0.01) M) | Ag(s)
Choose the correct statement
(A) The E_{cell} can be increased if concentration of $AgNO_3$ in half cell X is increased
(B) The E_{cell} will increase if concentration of $AgNO_3$ in cell Y is increased
(C) The E_{cell} will increase if concentration of $AgNO_3$ in cell X is decreased
(D) The E_{cell} will increase if concentration of $AgNO_3$ in cell Y is decreased

22. For a sample containing O_2 and SO_2 mark out the correct statement(s):
(A) At same temperature average speed of O_2 is greater than that of SO_2.
(B) If absolute temperature of O_2 is half to that of SO_2 gas than the speed distribution curves will be different.
(C) At same temperature & at lower speed, fraction of molecules of SO_2 will be greater than that of O_2.
(D) At same temperature & at higher speed, fraction of molecules of O_2 will be greater than that of SO_2.

23. Which among the following have square planar geometry?
(A) $[Ni(CO)_4]$ (B) $[Cu(NH_3)_4]^{2+}$
(C) $[Pt(NH_3)_2Cl_2]$ (D) $[Ni(CN)_4]^{2-}$

Space for Rough work

24. 1 mol of NH_3 gas (γ=1.33) at 27° C is expanded reversibly under adiabatic conditions to increase volume 8 times the initial volume. Correct option(s) is/are
(A) final temp. = 150 K (B) work done = –400 cal
(C) final temp. = 250 K (D) work done = –900 cal

25. Which of the following statement is correct about the compound of Xe?
(A) 3 mole XeF_4 on hydrolysis gives 2 mole XeO_3 and 1 mole Xe
(B) $XeO_3 + XeOF_4 \longrightarrow$ product (having see-saw shape)
(C) with RbF , both XeF_4 and XeF_6 act as Lewis acid
(D) with SbF_5 , both XeF_2 and XeF_4 act as Lewis base

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 26, 27 and 28 by appropriately matching the information given in the three columns of the following table.

Column – 1 (compound)	Column – 2 (test)	Column – 3 (properties)
(I) $(CH_3)_2$ CH – OH	(i) +ve iodoform test	(P) strongest C – O bond
(II) $CH_3 – CH_2 – OH$	(ii) turbidity with Lucas reagent in 2 min.	(Q) evolves CO_2 on oxidation followed by heating with soda-lime
(III) $(CH_3)_3$ C – OH	(iii) violet colouration with neutral $FeCl_3$	(R) most acidic
(IV) (p-NO_2) Ph – OH	(iv) blue colour in Victor – Meyer test	(S) least acidic

26. Correct match among the following:
(A) (I) (iv) (Q) (B) (IV) (iii) (R)
(C) (II) (i) (S) (D) (III) (ii) (P)

27. Correct match among the following:
(A) (I) (iii) (R) (B) (IV) (i) (Q)
(C) (II) (iv) (P) (D) (III) (ii) (S)

Space for Rough work

28. Correct match among the following:
(A) (I) (iv) (P) (B) (IV) (ii) (S)
(C) (II) (i) (Q) (D) (III) (iii) (R)

Answer questions 29, 30 and 31 by appropriately matching the information given in the three columns of the following table.

Column – 1 (cation)	Column – 2 (group reagent)	Column – 3 (appearence)
(I) Fe^{+3}	(i) H_2S in presence of HCl	(P) reddish brown precipitate with $K_4[Fe(CN)_6]$
(II) Ni^{+2}	(ii) NH_4Cl in presence of NH_4OH	(Q) blue precipitate with $K_4[Fe(CN)_6]$
(III) Cu^{+2}	(iii) $(NH_4)_2CO_3$ in presence of NH_4Cl and NH_4OH	(R) red precipitate with dimethyl glyoxime in alkaline medium
(IV) Ba^{+2}	(iv) H_2S in presence of NH_4OH	(S) apple green colour in Bunsen flame

29. Correct match among the following:
(A) (I) (i) (P) (B) (IV) (ii) (S)
(C) (II) (iv) (R) (D) (III) (iii) (Q)

30. Correct match among the following:
(A) (I) (iii) (S) (B) (IV) (ii) (R)
(C) (II) (iv) (Q) (D) (III) (i) (P)

31. Correct match among the following:
(A) (I) (ii) (Q) (B) (IV) (iii) (P)
(C) (II) (i) (R) (D) (III) (iv) (S)

Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. A 0.001 molal solution of a complex $[MA_8]$ in water has the freezing point of –0.0054°C. Assuming 100% ionization of the complex salt and K_f for H_2O = 1.86 km^{-1}, what is the ratio of observed and expected osmotic pressure?

33. How many number of maxima's in radial probability function curve of a 5p orbital?

34. In Hall – Heroult process, to obtain 4 kg of aluminium, how much amount (in kg) of graphite is approximately consumed?

35. How many moles NH_3 must be added to 2.0 litre of 0.80 M $AgNO_3$ in order to reduce the Ag^+ concentration to 5×10^{-8} M. (Given : K_f of $[Ag(NH_3)_2^+] = 10^8$)

36. Glycine, Alanine and Phenyl alanine on combining can form how many dipeptides?

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Let f : R $\to$ R, such that $f''(x) - 2f'(x) + f(x) = 2e^x$ and $f'(x) > 0 \ \forall$ x $\in$ R, then which of the following can be correct

(A) $|f(x)| = -f(x), \forall x \in R$ (B) $|f(x)| = f(x), \forall x \in R$

(C) $f(3) = -5$ (D) f(3) = 7

38. $(\vec{a} \times \vec{b}).(\vec{c} \times \vec{d})$ is equal to

(A) $\vec{a}\{\vec{b} \times (\vec{c} \times \vec{d})\}$ (B) $\{(\vec{a} \times \vec{b}) \times \vec{c}\}.\vec{d}$

(C) $(\vec{a}.\vec{c})(\vec{b}.\vec{d}) - (\vec{a}.\vec{d})(\vec{b}.\vec{c})$ (D) $(\vec{d} \times \vec{c}).(\vec{b} \times \vec{a})$

39. Let $|f(x)| \le \sin^2 x, \forall \ x \in R$, then

(A) f(x) is continuous at x = 0

(B) f(x) is differentiable at x = 0

(C) f(x) is continuous but not differentiable at x = 0

(D) f(0) = 0

40. If the reflection of the line $\bar{a}z + a\bar{z} = 0$ in the real axis is $\bar{\alpha}z + \alpha\bar{z} = 0$ in the simplest form then

(A) $\alpha + a$ is purely real (B) $\frac{\bar{\alpha}}{a} = \frac{\alpha}{\bar{a}}$

(C) $\bar{\alpha}\,\bar{a} = a\alpha$ (D) $\bar{\alpha} - \bar{a}$ is purely imaginary

Space for Rough work

41. Let A = $[a_{ij}]$ where $a_{ij} = u_i v_j$ $1 \le i \le n$ and $i \le j \le n$ $u_i, v_j \in R$ satisfies $A^5 = 16A$ then $T_r(A)$ is $(T_r \equiv \text{Trace of matrix})$

(A) 2 (B) 4
(C) –2 (D) –4

42. If $\left(1 + ax + bx^2\right)^4 = a_0 + a_1 x + a_2 x^2 + ... + a_8 x^8$ where $a, b, a_0, a_1, a_2 ..... a_8 \in R$ such that $a_0 + a_1 + a_2 \neq 0$ and $\begin{vmatrix} a_0 & a_1 & a_2 \\ a_1 & a_1 & a_0 \\ a_2 & a_0 & a_1 \end{vmatrix} = 0$ then

(A) $a_0 = a_1 \neq a_2$ (B) $a_1 = a_2 = 1$
(C) $a_0 = a_1 = a_2 = 4a$ (D) $b = \dfrac{5}{32}$

43. A parabola touches the bisectors of the angle obtained by the lines $x + 2y + 3 = 0$ and $2x + y + 3 = 0$ at the point (1, 1) and (0, -2). then

(A) The focus of parabola is $\left(\dfrac{1}{5}, \dfrac{-7}{5}\right)$
(B) The focus of parabola is (-2, 3)
(C) The equation of directrix is $3x + y + 4 = 0$
(D) The length of latus rectum is $\dfrac{16\sqrt{5}}{2^{\frac{3}{2}}}$

Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 44, 45 and 46 by appropriately matching the information given in the three columns of the following table.

$f(x) = x + \log_e x - x\log_e x,\ x \in (0, \infty)$.

List I contains information about zero of f(x), f'(x) and f''(x).
List II contains information about the limiting behavior of f(x), f'(x) and f''(x) at infinity
List III contains information about increasing/decrasing nature of f(x) and f'(x)

Column – 1		Column – 2		Column - 3	
(I)	$f(x) = 0$ for some $x \in (1, e^2)$	(i)	$\lim_{x\to\infty} f(x) = 0$	(P)	f is increasing in (0, 1)
(II)	$f'(x) = 0$ for some $x \in (1, e)$	(ii)	$\lim_{x\to\infty} f(x) = -\infty$	(Q)	f is decreasing in (e, e^2)
(III)	$f'(x) = 0$ for some $x \in (0, 1)$	(iii)	$\lim_{x\to\infty} f'(x) = 0$	(R)	f' is increasing in (0, 1)
(IV)	$f''(x) = 0$ for some $x \in (1, e)$	(iv)	$\lim_{x\to\infty} f''(x) = 0$	(S)	f' is decreasing in (e, e^2)

44. Which of the following option is the only CORRECT combination?
(A) (IV) (i) (S) (B) (I) (ii) (R)
(C) (III) (iv) (P) (D) (II) (iii) (S)

45. Which of the following option is the only CORRECT combination?
(A) (III) (iii) (R) (B) (I) (i) (R)
(C) (IV) (iv) (S) (D) (II) (ii) (Q)

46. Which of the following option is the only INCORRECT combination?
(A) (II) (iii) (P) (B) (II) (iv) (Q)
(C) (I) (iii) (P) (D) (III) (i) (R)

Space for Rough work

Answer questions 47, 48 and 49 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)	Minimum value of $\frac{1}{3}\left[\lvert z\rvert^2+\lvert z-3\rvert^2+\lvert z-6i\rvert^2\right]$	(i)	$M=1$	(P)	$\lvert z-2-3i\rvert^2+\lvert z-4-3i\rvert^2=M$ represent the equation of circle with least radius
(II)	If $\lvert z-2-2i\rvert\le 1$. Then minimum value of $\frac{1}{2}\lvert 2iz+4\rvert=M$	(ii)	$M=4$	(Q)	$\lvert z-2-3i\rvert+\lvert z+2+i\rvert=M$ represent an ellipse
(III)	$\lvert z+2-i\rvert=5$, Maximum value of $\frac{1}{5}\lvert 3z+9-7i\rvert=M$	(iii)	$M=10$	(R)	$\frac{\lvert z-z_1\rvert}{\lvert z-z_2\rvert}=M$ represent equation of bisector
(IV)	If $\left\lvert z-\frac{4}{z}\right\rvert=2$ Maximum value of $\left[\lvert z\rvert\right]=M$ (where [.] denotes the greatest integer function)	(iv)	$M=3$	(S)	$\omega(1+\omega)^M+\omega^2\left(\omega+\omega^2\right)^M+(-1)^M=0$ ω is cube root unity

47. Which of the following option is the only CORRECT combination?
(A) (I) (ii) (S) (B) (I) (iii) (P)
(C) (I) (iii) (Q) (D) (I) (ii) (Q)

48. Which of the following option is the only CORRECT combination?
(A) (II) (ii) (P) (B) (II) (i) (R)
(C) (II) (i) (S) (D) (II) (iv) (R)

49. Which of the following option is the only INCORRECT combination?
(A) (III) (ii) (S) (B) (III) (iii) (R)
(C) (III) (i) (P) (D) (III) (ii) (P)

Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. If two pts P and Q on the hyperbola $\frac{x^2}{9}-\frac{y^2}{16}=1$ whose centre is C be such that CP is perpendicular to CQ and a < b, then $\frac{1}{CP^2}+\frac{1}{CQ^2}=\frac{\alpha}{\beta}$ (α, β have no common divisor) then $\alpha+\beta-150=$ _____.

51. Two numbers of integral points on the hyperbola $x^2-y^2=(2000)^2$ is α. (An integral points is a point both of whose coordinates are integer). Then $\frac{\alpha}{49}=$ _____.

52. The graph of $x^2+y^2+6x-24y+72=0$ and $x^2-y^2+6x+16y-46=0$ intersect at four points. Sum of the distance of these four points from the point (-3, 2) be α, then $\frac{\alpha}{20}=$ _____.

53. Let $(x^3 + px^2 + 2x – 5)^{19} (x^2 + qx – 41)^8 (x^4 – x^3 + x – 7)^6 = x^{97} + 391x^{96} + a_{95}x^{95} + a_{94}x^{94} + ... a_1x + a_0$ be an identity, where p, q, $a_{95}....a_0$ are integers. Compute the smallest positive value of p.

54. $f(x + 1) = (-1)^{x+1} x – 2f(x)$ for $x \in N$ and $f(1) = f(1986)$. Compute the sum of digit of numbers $(f(1) + f(2) +..... f(1985))$

Space for Rough work